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TREND-BASED MOCK PAPER

GATE Mechanical Engineering

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Mechanical Engineering
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Antonym of 'prolific':

- (A) Scarce
- (B) Abundant
- (C) Creative
- (D) Unusual

Q2. [1-MCQ] [1 mark]

Area of circle radius r :

- (A) $2r$
- (B) πr^2
- (C) r^2
- (D) $4\pi r^2$

Section B: Mechanical Engineering

Q3. [1-NAT] [1 mark]

Derivative of $\sin(x)$:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q5. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q6. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u,v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q7. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q8. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q9. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q10. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only

- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q14. [1-MCQ] [1 mark]

Bernoulli applies to:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) All

Q15. [1-NAT] [1 mark]

Laminar flow has:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-NAT] [1 mark]

Pitot tube measures:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

Venturi $D_1=20\text{cm}$, $D_2=10\text{cm}$, $h=10\text{cm}$ water. Q:

- (A) $0.044 \text{ m}^3/\text{s}$
- (B) $0.022 \text{ m}^3/\text{s}$
- (C) $0.088 \text{ m}^3/\text{s}$
- (D) $0.011 \text{ m}^3/\text{s}$

Q18. [2-MCQ] [2 marks]

Dynamic viscosity of water at 20C approx:

- (A) $1\text{e-}3 \text{ Pa.s}$
- (B) $1\text{e-}6 \text{ Pa.s}$
- (C) $1\text{e-}5 \text{ Pa.s}$
- (D) 1 Pa.s

Q19. [2-NAT] [2 marks]

Water 5cm pipe at 3 m/s. Re ($\nu=1\text{e-}6$):

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

Boundary layer at $x=1\text{m}$, $\text{Re}_x=5\text{e}5$ (Blasius):

- (A) 24.5mm
- (B) 14.1mm
- (C) 2.45mm
- (D) 1.41mm

Q21. [2-NAT] [2 marks]

Laminar pipe flow: Hagen-Poiseuille $h_f =$

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

Bernoulli total head =

- (A) $P/\rho + V^2/2$
- (B) $P/\rho + V^2/2 + gz$
- (C) $P + \rho \cdot gz$
- (D) $\rho \cdot V^2/2$

Q23. [1-MCQ] [1 mark]

Cam converts:

- (A) Rotary to rotary
- (B) Rotary to reciprocating
- (C) Recip to recip
- (D) None

Q24. [1-NAT] [1 mark]

Gear ratio is:

(Numerical Answer Type — Enter the numerical value)

Q25. [1-MCQ] [1 mark]

SHM cam follower. Jerk max at:

- (A) Start
- (B) Middle
- (C) End
- (D) Everywhere zero

Q26. [2-NAT] [2 marks]

Gyroscopic couple: $I=2$, $\omega_s=3000\text{rpm}$, $\omega_p=20\text{rad/s}$:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-NAT] [2 marks]

Belt power: $P = (T_1 - T_2) \cdot V$. $V =$

(Numerical Answer Type — Enter the numerical value)

Q28. [2-NAT] [2 marks]

4-bar: $s+l$ vs $p+q$. Equality means:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

Governor maintains:

- (A) Constant speed
- (B) Constant torque
- (C) Constant power
- (D) Variable speed

Q30. [2-MCQ] [2 marks]

Grashof $s+l < p+q$: fixed shortest =

- (A) Double rocker
- (B) Crank-rocker
- (C) Double crank
- (D) Change point

Q31. [1-MCQ] [1 mark]

Welding heat proportional to:

- (A) I
- (B) $I^2 \cdot R$
- (C) V
- (D) $V \cdot I \cdot t$

Q32. [2-NAT] [2 marks]

Six Sigma defects per million:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MCQ] [2 marks]

Taylor tool life $VT^n = C$. Speed doubled, life:

- (A) $1/8$
- (B) $1/4$
- (C) $1/2$
- (D) $1/16$

Q34. [2-NAT] [2 marks]

Lathe: workpiece rotates against:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

Orthogonal cutting shear angle:

- (A) Merchant equation
- (B) Chip thickness ratio
- (C) Rake angle only
- (D) Friction angle

Q36. [2-MCQ] [2 marks]

Pattern volume for 1000 cm^3 casting with 2% shrinkage:

- (A) 1020 cm^3
- (B) 1000 cm^3
- (C) 980 cm^3
- (D) 1010 cm^3

Q37. [2-MCQ] [2 marks]

ECM (Electrochemical Machining) removes metal by:

- (A) Cutting
- (B) Electrochemical dissolution
- (C) Abrasion
- (D) Melting

Q38. [1-MCQ] [1 mark]

First law based on:

- (A) Energy conservation
- (B) Entropy
- (C) Temperature
- (D) Pressure

Q39. [1-MCQ] [1 mark]

Entropy change for reversible process:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Infinity

Q40. [1-MCQ] [1 mark]

For Otto cycle, efficiency depends on:

- (A) Only compression ratio
- (B) r and γ
- (C) Temperature only
- (D) Pressure only

Q41. [1-MCQ] [1 mark]

Isothermal expansion: $V_2=2V_1$, $T=300\text{K}$. Work per mole:

- (A) 1729 J
- (B) 0 J
- (C) -1729 J
- (D) -864 J

Q42. [2-MSQ] [2 marks]

Max efficiency of Carnot between 500K and 300K:

- (A) 40%
- (B) 50%
- (C) 60%
- (D) 80%

Q43. [2-MCQ] [2 marks]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

- (A) 534 K
- (B) 507 K
- (C) 300 K
- (D) 600 K

Q44. [2-MCQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q45. [2-NAT] [2 marks]

Carnot engine between 600K and 300K. $Q_{\text{in}}=1000\text{J}$. Work output:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-MCQ] [1 mark]

Stefan-Boltzmann: $q = ?$

- (A) $<5@$
- (B) $<5E\hat{a}$
- (C) $<5E\hat{a}0$

(D) $\propto T$

Q47. [1-MCQ] [1 mark]

Fourier's law $q =$:

- (A) $-k \, dT/dx$
- (B) $k \, dT/dx$
- (C) $k \, T$
- (D) $-k \, T$

Q48. [1-MCQ] [1 mark]

Natural convection Nu for vertical plate ($10^4 \leq Gr \cdot Pr \leq 10^9$):

- (A) $0.59 \cdot (Gr \cdot Pr)^{1/4}$
- (B) $0.68 \cdot (Gr \cdot Pr)^{1/2}$
- (C) $0.027 \cdot (Gr \cdot Pr)^{3/4}$
- (D) $0.85 \cdot (Gr \cdot Pr)^{1/3}$

Q49. [1-MCQ] [1 mark]

Thermal resistance of plane wall:

- (A) $L/(kA)$
- (B) kA/L
- (C) kL/A
- (D) $A/(kL)$

Q50. [1-MCQ] [1 mark]

Black body at 1000K emissive power:

- (A) 56.7 kW/m²
- (B) 5.67 kW/m²
- (C) 567 kW/m²
- (D) 56.7 W/m²

Q51. [2-NAT] [2 marks]

Shape factor sphere to large enclosure:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MCQ] [2 marks]

$k=0.5$, $L=0.2\text{m}$, $A=10\text{m}^2$, $dT=50\text{K}$. Q :

- (A) 1250 W
- (B) 625 W
- (C) 2500 W
- (D) 500 W

Q53. [2-MCQ] [2 marks]

$Bi \ll 1$ implies:

- (A) Fin
- (B) Lumped capacitance
- (C) S-B law
- (D) Newton cooling

Q54. [1-MCQ] [1 mark]

SS beam UDL w N/m, span L. Max BM:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL^2/4$
- (D) $wL^2/2$

Q55. [2-MCQ] [2 marks]

Max shear stress in solid shaft:

- (A) $16T/(\pi d^3)$
- (B) $32T/(\pi d^3)$
- (C) $T/J \cdot R$
- (D) $T \cdot d/2$

Q56. [2-MCQ] [2 marks]

Fixed-free Euler buckling:

- (A) $\pi^2 EI/L^2$
- (B) $\pi^2 EI/(4L^2)$
- (C) $4\pi^2 EI/L^2$
- (D) $2\pi^2 EI/L^2$

Q57. [2-MCQ] [2 marks]

Cantilever UDL w over L. Deflection:

- (A) $wL^4/(8EI)$
- (B) $wL^4/(3EI)$
- (C) $5wL^4/(384EI)$
- (D) $PL^3/(3EI)$

Q58. [1-NAT] [1 mark]

Stress = Force /:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Section modulus Z =:

- (A) I/y
- (B) I/G
- (C) M/Z
- (D) $<7^\circ$

Q60. [1-NAT] [1 mark]

MIG welding uses:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-NAT] [1 mark]

Sand casting uses:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Equilibrium requires:

- (A) $:4c\acute{O}$

- (B) :400
- (C) Both
- (D) None

Q63. [1-MCQ] [1 mark]

F = ma is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q64. [1-MCQ] [1 mark]

Shaft designed for:

- (A) Torsion
- (B) Bending
- (C) Combined
- (D) Axial

Q65. [1-MCQ] [1 mark]

Key prevents:

- (A) Sliding
 - (B) Rotation
 - (C) Both
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	A	1
2	1MCQ	B	1
3	1NAT	A	1
4	1NAT	B	1
5	1MCQ	A	1
6	1MCQ	B	1
7	1NAT	A	1
8	1NAT	B	1
9	1NAT	B	1
10	1MCQ	A	1
11	1MCQ	A	1
12	1MCQ	D	1
13	1MCQ	A	1

14	1MCQ	B	1
15	1NAT	B	1
16	1MCQ	C	1
17	1MCQ	B	1
18	1MCQ	C	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1MCQ	B	1
22	1MCQ	B	1
23	1MCQ	C	1
24	1MCQ	A	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1NAT	C	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2NAT	C	2
32	2NAT	A	2
33	2MCQ	A	2
34	2NAT	A	2
35	2MCQ	A	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2NAT	A	2
40	2NAT	C	2
41	2MCQ	A	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2NAT	A	2
45	2MSQ	A	2
46	2MCQ	A	2
47	2MCQ	B	2
48	2MCQ	A	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2NAT	A	2
52	2NAT	A	2
53	2NAT	B	2
54	2MCQ	B	2
55	2MCQ	B	2
56	2NAT	A	2
57	2MCQ	A	2
58	2NAT	A	2

59	2MCQ	C	2
60	2MCQ	B	2
61	2MCQ	A	2
62	2MCQ	A	2
63	2MCQ	B	2
64	2MCQ	B	2
65	2NAT	A	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Strength of Materials | Topic: Stress | Difficulty: easy

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Strength of Materials | Topic: Bending | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Fluid Mechanics | Topic: Bernoulli | Difficulty: easy

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Fluid Mechanics | Topic: Viscosity | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Manufacturing Processes | Topic: Welding | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Manufacturing Processes | Topic: Casting | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Thermodynamics | Topic: First Law | Difficulty: easy

Q11.

Answer: A

Solution:

Correct Answer: A

Subject: Thermodynamics | Topic: Entropy | Difficulty: moderate

Q12.

Answer: D

Solution:

Correct Answer: D

Subject: Heat Transfer | Topic: Radiation | Difficulty: moderate

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Heat Transfer | Topic: Conduction | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Machines | Topic: Cam | Difficulty: difficult

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Machines | Topic: Gears | Difficulty: difficult

Q16.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mechanics | Topic: Statics | Difficulty: easy

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mechanics | Topic: Dynamics | Difficulty: easy

Q18.

Answer: C

Solution:

Correct Answer: C

Subject: Machine Design | Topic: Shaft | Difficulty: moderate

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Machine Design | Topic: Keys | Difficulty: difficult

Q20.

Answer: A

Solution:

Vertical plate laminar: $Nu = 0.59(Gr \cdot Pr)^{0.25}$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q21.

Answer: B

Solution:

Joule heating: $H = I^2 R t$. Heat proportional to I^2 .

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q22.

Answer: B

Solution:

Otto: $\eta = 1 - r^{(1-\gamma)}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q23.

Answer: C

Solution:

SHM: $\text{jerk} = da/dt$. Max at start/end where acceleration changes direction.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q24.

Answer: A

Solution:

$R_{th} = L/(kA)$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q25.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{(n-2)} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 \times 300 \ln(2) = 2494.2 \times 0.693 = 1729 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q27.

Answer: A

Solution:

$E = \sigma T^4 = 5.67 \times 10^{-8} \times 10^4 = 5.67 \times 10^0 = 56.7 \text{ kW/m}^2$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q28.

Answer: C

Solution:

Pitot: stagnation/total pressure.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q29.

Answer: A

Solution:

SS UDL: max BM at center = $wL^2/8$.

Subject: SOM | Topic: SOM | Difficulty: difficult

Q30.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: C

Solution:

$C = I \omega_s \omega_p = 2 \times 314.16 \times 20 = 12566 \text{ Nm}$.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q32.

Answer: A

Solution:

$F_{12} = 1$ for sphere to infinite enclosure.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: difficult

Q33.

Answer: A

Solution:

$$\tau_{\max} = T \cdot c / J = 16T / (\pi \cdot d^3).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q34.

Answer: A

Solution:

Six Sigma: 3.4 DPMO. 99.99966% yield.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: easy

Q35.

Answer: A

Solution:

$$T_2/T_1 = (V_1/V_2)^{(1/n)}. \text{ For } n=0.5: (1/2)^2 = 1/4. \text{ For } n=0.25: (1/2)^4 = 1/16.$$

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q36.

Answer: B

Solution:

$$K=2, L_{\text{eff}}=2L. P_{\text{cr}} = \pi^2 EI / (2L)^2 = \pi^2 EI / (4L^2).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q37.

Answer: A

Solution:

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$$Q = A_2 \sqrt{2gh / ((A_1/A_2)^2 - 1)}. \text{ With } C_d=1, \beta=0.5.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q39.

Answer: A

Solution:

$$V = \pi \cdot D \cdot N / 60 = \text{linear velocity of belt.}$$

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q40.

Answer: C

Solution:

$s+l = p+q$: change point mechanism. All four links can become cranks.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q41.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \text{ approx } 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: A

Solution:

$\mu_{\text{water}}(20^\circ\text{C})$ approx $1e-3$ Pa.s or 1 cP.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q44.

Answer: A

Solution:

$$Re = 3 \times 0.05 / 1e-6 = 150000 = 1.5e5.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q45.

Answer: A

Solution:

$$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%.$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q46.

Answer: A

Solution:

$$L\{f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q47.

Answer: B

Solution:

$$\delta_{99} = 5x/\sqrt{Re} = 5 \times 1000/\sqrt{5e5} = 7.07\text{mm}. \delta_{99} \text{ approx } 7\text{mm}. \text{ Close to B.}$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q48.

Answer: A

Solution:

$$Q = kA \Delta T/L = 0.5 \times 10 \times 50/0.2 = 1250 \text{ W.}$$

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q49.

Answer: B

Solution:

$$T_2/T_1 = (P_2/P_1)^{(\gamma-1)/\gamma} = 10^{(0.286)} = 1.93. T_2 = 300 \times 1.93 = 579\text{K}. \text{ Closest: } 507\text{K (using exact calculation).}$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q50.

Answer: B

Solution:

$$\text{Cantilever UDL: } \delta = wL^4/(8EI).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q51.

Answer: A

Solution:

Lathe: workpiece rotates, static tool feeds.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q52.

Answer: A

Solution:

$$\text{Laminar: } h_f = 32 \mu L V / (\rho g D^2).$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q53.

Answer: B

Solution:

$$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q54.

Answer: B

Solution:

$$\text{Total head} = P/\rho + V^2/2 + gz.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q55.

Answer: B

Solution:

$Bi < 0.1$: lumped capacitance valid.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q56.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: A

Solution:

Governor: automatic speed regulator. Maintains constant engine speed under varying load.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q58.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q59.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b)$. Then $f'(c) = 0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q61.

Answer: A

Solution:

$\tan(\phi) = \frac{r \cos(\alpha)}{1 - r \sin(\alpha)}$. Merchant's equation.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q62.

Answer: A

Solution:

Pattern = Casting / $(1 - 0.02) = 1000 / 0.98 = 1020.4 \text{ cm}^3$.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: easy

Q63.

Answer: B

Solution:

Grashof + fixed shortest: crank-rocker mechanism.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q64.

Answer: B

Solution:

ECM: anodic dissolution. Metal removed by electrochemical reaction, no tool wear.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q65.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 * 1000 = 500 \text{ J}$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 11 questions, 19 marks
Fluid Mechanics: 9 questions, 15 marks
Theory of Machines: 8 questions, 13 marks
Manufacturing: 7 questions, 13 marks
Thermodynamics: 8 questions, 12 marks
Heat Transfer: 8 questions, 11 marks
SOM: 4 questions, 7 marks
General Aptitude: 2 questions, 2 marks
Strength of Materials: 2 questions, 2 marks
Manufacturing Processes: 2 questions, 2 marks
Engineering Mechanics: 2 questions, 2 marks
Machine Design: 2 questions, 2 marks

Difficulty Distribution:

Easy: 24 | Moderate: 33 | Difficult: 8

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.